

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation **COURSE CODE:** CSA13

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:25 Duration : 1 Hour

Analytical Problem

Code of Conduct:

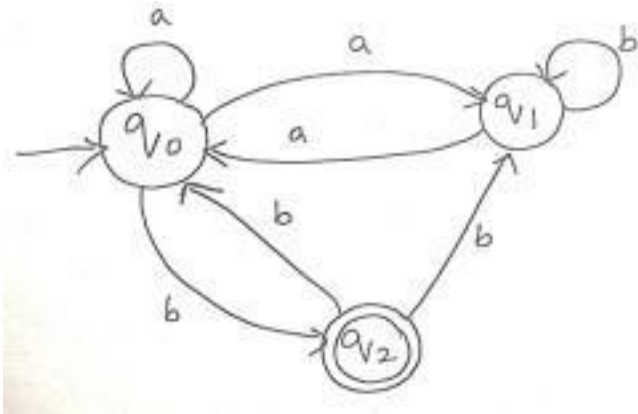
I M.VINAY(192311146) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.VINAY

SIMATS ENGINEERING ASSESSMENT TOOL

1. Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{a, b\}$. Construct a DFA that represents the same language L .

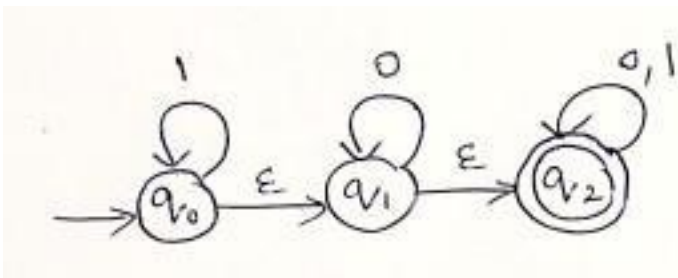


2. Let L be the language of words of length at least 2 whose last two letters are different from each other. For example $abaab$ is in L but not bb . Construct an NFA for L . Then draw the equivalent DFA.

3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string **01011** is accepted by the NFA.

4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet $\Sigma = \{0, 1\}$. Construct a DFA that represents the same language.

5. A NFA with ϵ -transitions is given below. Find ϵ -closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states.



SIMATS ENGINEERING ASSESSMENT TOOL

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
		m			

Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Answer

To solve **Question 1**, you need to convert the given **NFA into an equivalent DFA** using the **subset construction (powerset) method**. The NFA shown on page 2 has three states (q_0 , q_1 , q_2), with q_2 as the accepting state.

Ans = Read the NFA

From the diagram:

- **Start state:** q_0
- **Final state:** q_2

Transitions:

State	a	b
q_0	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
q_2		$\{q_0, q_1\}$

$\emptyset$

Start the DFA

The DFA start state is the set containing the NFA start state:

$A = \{q_0\}$

Step 3: Find New States State A = $\{q_0\}$

- $a \rightarrow \{q_0, q_1\} = B$
- $b \rightarrow \{q_2\} = C$

State B = $\{q_0, q_1\}$

Combine transitions of q_0 and q_1 .

On **a**:

- $q_0 \rightarrow \{q_0, q_1\}$
- $q_1 \rightarrow \{q_0\}$ Union:

$\{q_0, q_1\}$

So B

--a--> B

On **b**:

- $q_0 \rightarrow \{q_2\}$
- $q_1 \rightarrow \{q_1\}$ Union:

$\{q_1, q_2\} = D$

State C = $\{q_2\}$

On **a**

$\emptyset$

Dead state =

E On **b** $q_2 \rightarrow$

$\{q_0, q_1\}$ So

C --b--> B

State D = {q1,q2}

On **a**

- $q1 \rightarrow q0$
- $q2 \rightarrow \emptyset$

Result

$\{q0\} = A$

On **b**

- $q1 \rightarrow q1$
- $q2 \rightarrow \{q0, q1\}$

Union

$\{q0, q1\} = B$

State E = Dead State ($\emptyset$) Both
inputs remain in E.

DFA Transition Table

DFA State	NFA States	a	b
	$\{q0\}$ B C		
	$\{q0, q1\}$ B D		
	$\{q2\}$ E B		
	$\{q1, q2\}$ A B		
	$\emptyset$ E E		

Step 5: Final States

A DFA state is accepting if it contains **q2**.

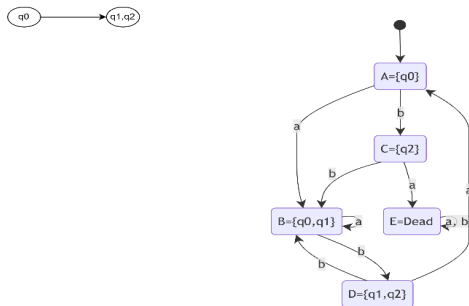
Therefore, Final

States = C , D

because $C = \{q2\}$

$D = \{q1, q2\}$

Final DFA Diagram (Exam)



Transitions:

- A --b--> C
 B --b--> D
 C --b--> B
 D --a--> A D --b--> B

Question 2:

1. Design an NFA for the language.
2. Convert the NFA into an equivalent DFA.

Ans = Given Language

L = { Strings of length at least 2 whose last two letters are different }

Accepted strings:

- ab
- ba
- aab
- abaab

Rejected strings:

- aa
- bb
- abbaa (last two letters are aa)
- abbbb (last two letters are bb)

Step 1: Design the NFA

States

- **q0** = Start
- **q1** = Last symbol is a
- **q2** = Last symbol is b
- **q3** = Accept (ab)
- **q4** = Accept (ba)

NFA Transition Table State a b

→ q0 {q0,q1} {q0,q2}

q1 ∅ {q3}

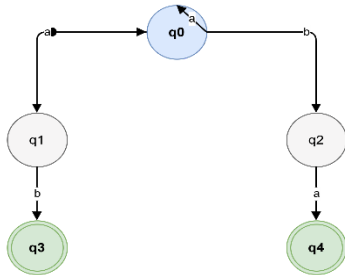
q2 {q4} ∅

★ q3 ∅ ∅

★ q4 ∅ ∅

(★ = Final State)

NFA Diagram



q3 accepts strings ending with **ab**.

- **q4** accepts strings ending with **ba**.

Step 2: Convert NFA to DFA DFA States

DFA State	NFA States
A	{q0}
B	{q0,q1}
C	{q0,q2}
D	{q0,q2,q3}
E	{q0,q1,q4}

DFA Transition Table

DFA State a b

A={q0} B C

B={q0,q1} B D

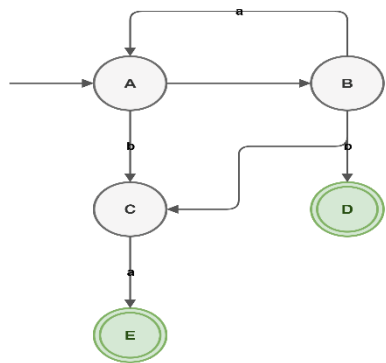
C={q0,q2} E C

★ D={q0,q2,q3} E C

★ E={q0,q1,q4} B D Accepting states:

- **D**
- **E**

DFA Diagram



Question 3

Design a Nondeterministic Finite Automaton (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string 01011 is accepted by the NFA.

Formal Definition

$[M=(Q,\Sigma,\delta,q_0,F)]$

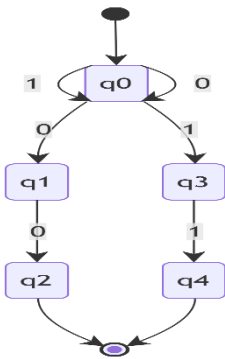
Where:

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0,1\}$
- q_0 = Start State
- $F = \{q_2, q_4\}$

Transition Function

Current State	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0, q_3\}$
q_1	$\{q_2\}$	
$* q_2$		$\emptyset$
q_3	$\emptyset$	$\{q_4\}$
$* q_4$	$\emptyset$	

Step 3: NFA Diagram



- q_2 accepts strings ending in 00.
- q_4 accepts strings ending in 11.

Step 4: Check the String 01011

Input:

01011

One accepting path is:

$q_0 \xrightarrow{0} q_0$

$q_0 \xrightarrow{1} q_0$

$q_0 \xrightarrow{0} q_0$

$q_0 \xrightarrow{1} q_3$

$q_3 \xrightarrow{1} q_4$

Since the computation ends in q_4 , which is a final state:

01011 is Accepted

(The last two bits are **11**.)

Question 4,

convert the given NFA into an equivalent DFA using the subset construction method. The NFA accepts all binary strings that contain **001** as a substring.

Ans = NFA Transition Table

From the NFA:

State	0	1	$\emptyset$
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0\}$	
q_1	$\{q_2\}$		(q_3 is the accepting state.)
q_2		$\emptyset$	
		$\{q_3\}$	Construct DFA States
$* q_3$	$\{q_3\}$	$\{q_3\}$	Start state: $A = \{q_0\}$

Find transitions for each set.

A = {q0}

- $0 \rightarrow \{q_0, q_1\} = \mathbf{B}$
- $1 \rightarrow \{q_0\} = \mathbf{A}$

B = {q0, q1}

- $0 \rightarrow \{q_0, q_1, q_2\} = \mathbf{C}$
- $1 \rightarrow \{q_0\} = \mathbf{A}$

C = {q0, q1, q2}

- $0 \rightarrow \{q_0, q_1, q_2\} = \mathbf{C}$
- $1 \rightarrow \{q_0, q_3\} = \mathbf{D}$

D = {q0,q3}

- $0 \rightarrow \{q0,q1,q3\} = \mathbf{E}$
- $1 \rightarrow \{q0,q3\} = \mathbf{D}$

E = {q0,q1,q3}

- $0 \rightarrow \{q0,q1,q2,q3\} = \mathbf{F}$
- $1 \rightarrow \{q0,q3\} = \mathbf{D}$

F = {q0,q1,q2,q3}

- $0 \rightarrow \mathbf{F}$
- $1 \rightarrow \mathbf{D}$

DFA Transition Table

DFA State	NFA States	0	1
A	{q0}	B	A
B	{q0,q1}	C	A
C	{q0,q1,q2}	C	D
* D	{q0,q3}	E	D
* E	{q0,q1,q3}	F	D
* F	{q0,q1,q2,q3}	F	D

Final States

Any DFA state containing **q3** is a final state.

Therefore:

Final States = D, E, F

Question 5

1. Find the **ϵ -closure** of each state.
2. Construct the **equivalent DFA**.
3. Draw the **transition table**.
4. Mark the **start state** and **final state**.

Given ϵ -NFA

From the diagram:

- **q0** has a loop on **1**.
- **q0** $\xrightarrow{\epsilon}$ **q1**
- **q1** has a loop on **0**.
- **q1** $\xrightarrow{\epsilon}$ **q2** • **q2** has a loop on **0,1**.
- **q2** is the final state.

Find ϵ -Closures

ϵ -closure(q0) From q0: q0

-- $\epsilon \rightarrow$ q1 q1 -- $\epsilon \rightarrow$ q2 So,

ϵ -closure(q0) = {q0, q1, q2}

ϵ -closure(q1) From q1:

q1 -- $\epsilon \rightarrow$ q2 Therefore,

ϵ -closure(q1) = {q1, q2}

ϵ -closure(q2) No

ϵ -transition from q2.

Therefore,

ϵ -closure(q2) = {q2}

Step 3: Construct DFA DFA Start State

A = ϵ -closure(q0)

= {q0, q1, q2}

From A={q0,q1,q2}

On 0

- q0 $\rightarrow$ $\emptyset$
- q1 $\rightarrow$ q1
- q2 $\rightarrow$ q2 Move = {q1, q2} ϵ -closure({q1, q2})

= {q1, q2} Call it **B**.

On 1

- q0 $\rightarrow$ q0
- q1 $\rightarrow$ $\emptyset$
- q2 $\rightarrow$ q2 Move = {q0, q2} ϵ -closure({q0, q2})

= {q0, q1, q2}

= **A**

State B={q1,q2}

On 0

- q1 $\rightarrow$ q1
- q2 $\rightarrow$ q2

Result

{q1, q2} = **B**

On 1

- $q_1 \rightarrow \emptyset$
- $q_2 \rightarrow q_2$

Move

$\{q_2\}$

Call it **C**.

State C = $\{q_2\}$

On **0**

$\{q_2\}$

On **1**

$\{q_2\}$

Step 4: DFA Transition Table DFA State NFA States 0 1

★ A $\{q_0, q_1, q_2\}$ B A

★ B $\{q_1, q_2\}$ B C

★ C $\{q_2\}$ C C

Step 5: Final States

Since every DFA state contains **q_2** , which is the NFA's accepting state:

Final States = A, B, C